Matrix inequalities and norms

Sharp additive triangle constant for an odd number of contractions

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Sharpness for every odd summand count needs a new extremal construction or obstruction; the payoff is concentrated in operator triangle inequalities.

Resolution — 2026-09-11

Affirmative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS.**

The sharp additive contraction constant is $c_k = k/4$ for every $k \geq 2$, including every odd summand count. Exact dimension-two extremizers match the universal upper bound.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: complete proof PDF, standalone TeX, Theorem 1.1 and its proof; authorship and scope. The independent review checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. Submission record.

Problem statement

For each integer $k \geq 2$, let c_k be the infimum of all $c \geq 0$ such that

$$\left|\sum_{j=1}^{k} A_j\right| \preceq cI_n + \sum_{j=1}^{k} |A_j|$$

for every $n \geq 1$ and all $A_1, \dots, A_k \in \mathbb{C}^{n\times n}$ satisfying $\|A_j\|_2 \leq 1$. Here $|A| = (A^*A)^{1/2}$, $\|\cdot\|_2$ denotes the operator norm, and $X \preceq Y$ means $Y - X$ is positive semidefinite.

Is $c_k = k/4$ for every odd integer $k \geq 3$?

Why it matters

The scalar absolute-value triangle inequality fails in matrix order. The optimal additive correction quantifies that failure for uniformly bounded summands, in a form usable in positive block-matrix estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Diagonal and off-diagonal blocks of positive definite partitioned matrices*, arXiv:2307.02034v3 (15 December 2023), Corollary 4.4 and the conjecture immediately following Remark 4.5. Primary text.
2. E.-Y. Lee, *How to compare the absolute values of operator sums and the sums of absolute values?*, Operators and Matrices 6(3) (2012), 613–619; §2 supplies related triangle inequalities. Primary paper, DOI.

Status check — 2026-09-10

Corollary 4.4 establishes $c_k \leq k/4$ for every k and sharpness for even k; the source separately conjectures sharpness for all odd $k > 1$. Its latest version remains v3. Searches included *Bourin Lee contractions odd k constant k/4*, *three contractions 3/4 sharp conjecture*, and *Bourin contractions sharp 2025 2026*. The 2026 symmetric-modulus papers concern different inequalities. No resolution of this additive odd-summand problem was located.

Audit update (2026-09-10): Rechecked the conjecture after Remark 4.5 in Bourin–Lee and searched for odd-summand sharpness results. The proved even-summand statement is outside the target, which already restricts to odd k. This is a bounded literature check, not a proof that no solution exists.